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An AI-Powered Conversational Platform for Experience-Led Travel Discovery and Dynamic Itinerary Management

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Abstract—Traditional travel planning platforms operate on a transactional, search-led paradigm that requires users to predetermine their destination and dates, failing to capture abstract user intent or desired experiences. Additionally, these systems generate static itineraries that cannot adapt to the dynamic and unpredictable nature of real-world travel scenarios such as weather disruptions, overcrowding, or spontaneous preference changes. This paper introduces a novel hybrid AI system that addresses both discovery and execution phases of travel planning. Our solution comprises two primary innovations: (1) the AI Experience Weaver, a conversational Natural Language Processing (NLP) engine integrated with an experiential tagbased Knowledge Graph for generating personalized, multi-day itineraries from abstract user queries, and (2) the AI Travel Sync, a context-aware, proactive replanning mechanism that dynamically adjusts itineraries during trips based on real-time constraints. Built on the MERN stack with a curated MongoDB Knowledge Base enriched with semantic experiential tags, this system demonstrates a paradigm shift from destination-centric to experience-centric travel planning, offering both inspiration and intelligent adaptability throughout the travel lifecycle.

Index Terms—Natural Language Processing, Conversational AI, Experience-Based Recommender Systems, Knowledge Graphs, Context-Aware Systems, Dynamic Itinerary Planning, Travel Technology

I. Introduction

The global travel and tourism industry, valued at over $9 trillion annually, relies heavily on digital platforms for discovery, planning, and booking. However, current digital travel ecosystems are fundamentally constrained by their transactional architecture. These platforms function primarily as search engines, requiring users to input structured queries with predetermined parameters—destination, dates, budget—before presenting options. This "search-first" model creates critical limitations undermining user experience and market efficiency.

Modern travelers, particularly Millennials and Generation Z, increasingly seek experiences over destinations. A user desiring "a quiet, romantic escape in the mountains" possesses clear intent but lacks structured knowledge to formulate effective search queries. Current platforms cannot interpret such abstract, emotion-driven requests, forcing users into extensive manual research across travel blogs, social media, and review sites. This burden significantly increases cognitive load and planning friction.

Existing platforms also generate static itineraries that remain fixed post-booking. Real-world travel is inherently dynamic—subject to weather changes, site closures, transportation delays, or spontaneous preference shifts. When disruptions occur, travelers must manually research alternatives under stress, often sacrificing planned experiences. No mainstream platform provides intelligent, automated mechanisms to handle these contingencies.

Current recommender systems, while effective for suggesting individual components (hotels, flights), lack the generative capability to assemble complete, multi-day itineraries that holistically match abstract experiential intent. The absence of semantic understanding limits personalization depth.

A. Research Contributions

This paper addresses these interconnected challenges through an integrated AI system with two novel core mechanisms:

1) AI Experience Weaver: A conversational discovery engine employing lightweight NLP for intent extraction, coupled with a semantically enriched Knowledge Graph, to generate complete, personalized itineraries from abstract, experience-led natural language queries.
2) AI Travel Sync: A context-aware, rule-based replanning engine that monitors trip execution and proactively suggests viable alternatives when real-world constraints conflict with planned activities, enabling dynamic itinerary adaptation.

The proposed system represents a fundamental architectural shift: from reactive, search-led tools to proactive, intelligent travel companions that understand abstract intent, generate holistic plans, and adapt dynamically throughout the travel lifecycle.

II. Literature Survey

A. NLP in Conversational Systems

Jurafsky and Martin [1] demonstrate that modern NLP systems excel at extracting structured information from user queries through syntactic parsing and named entity recognition. However, contemporary conversational agents remain predominantly single-domain and task-oriented, struggling with ambiguous, abstract, or emotion-laden queries lacking concrete parameters. Zhang et al. [7] explore transformerbased models (BERT, GPT) for intent classification, demonstrating superior performance but identifying significant challenges: computational cost, large dataset requirements, and limited explainability for debugging.

B. Recommender Systems in Travel

Ricci et al. [2] validate the effectiveness of collaborative and content-based filtering for suggesting individual travel components. The core limitation identified is architectural: these systems are inherently destination-first, unable to construct complete multi-day itineraries when no destination is initially specified. Bobadilla et al. [9] provide a comprehensive survey of recommendation algorithms, emphasizing that current approaches excel at similarity-based matching but lack compositional reasoning capabilities required for complex, multi-component planning.

C. Knowledge Graph-Based Recommendations

Wang et al. [3] demonstrate that Knowledge Graphs enable multi-hop reasoning—traversing multiple relationships to find items matching complex criteria. KG-based systems provide more explainable recommendations than traditional approaches, though constructing high-quality graphs requires substantial manual curation. Their work establishes that semantic representations outperform vector embeddings for tasks requiring logical reasoning over relationships.

D. Constraint-Based Configuration Systems

Zanker et al. [11] explore constraint-based personalized configuration of product and service bundles, demonstrating that constraint satisfaction approaches can assemble complex multi-component offerings satisfying user requirements. However, their work focuses on static configuration rather than dynamic replanning, and assumes users can articulate constraints explicitly rather than through abstract experiential language. This highlights the need for systems combining semantic understanding with constraint-based assembly.

E. Context-Aware Recommender Systems

Adomavicius and Tuzhilin [4] establish that incorporating real-time contextual signals (location, time, weather) significantly improves recommendation relevance. Baltrunas et al. [10] apply these principles using matrix factorization with contextual features. The limitation is that implementations remain reactive—using context to filter recommendations rather than proactively restructuring plans.

F. Gamification for Engagement

Deterding et al. [8] validate that gamification elements significantly increase user engagement and loyalty in nongame applications. In travel contexts, they advocate for intrinsic gamification—rewards tightly coupled to core value propositions, such as encouraging discovery or rewarding adaptive behavior. This suggests potential for systems that incentivize users to explore AI-suggested alternatives rather than rigidly adhering to fixed plans.

G. Gap Analysis

No existing system integrates NLP-based experiential discovery with dynamic, constraint-aware replanning to address both the inspiration phase and execution phase of travel planning. Current platforms focus exclusively on transactional booking, while research prototypes explore individual components (NLP, recommendations, context-awareness) in isolation. The integration of these capabilities into a unified, end-to-end travel companion represents the primary contribution of this work.

III. Problem Statement

The contemporary digital travel ecosystem suffers from three fundamental deficiencies:

A. The Inspiration Gap

Current platforms mandate that users specify destination and dates before receiving recommendations. This fails users with abstract, experience-driven intent (“I want a peaceful escape”). The absence of NLP-powered intent extraction combined with semantically rich Knowledge Graphs means platforms cannot bridge the gap between abstract experiential concepts and concrete travel components.

B. The Static Itinerary Problem

All major platforms generate static itineraries that remain fixed post-planning. When disruptions occur, travelers lack automated assistance, forcing manual research under stress. The lack of context-aware, automated replanning mechanisms means platforms cannot function as intelligent co-pilots adapting plans dynamically to maintain trip quality.

C. The Generative Recommender Gap

Existing recommender systems excel at suggesting individual components but cannot compose complete, multi-day itineraries as cohesive wholes satisfying unified experiential themes. Current architectures lack integration of semantic Knowledge Graphs with generative algorithms capable of assembling logically structured, multi-day plans.

IV. Proposed Solution

A. System Architecture Overview

The Aura Travel prototype employs a four-layer hybrid architecture:

Layer 1: User Interface - A React.js web application presenting a conversational prompt: "What kind of experience are you dreaming of?" This signals that the system understands abstract queries rather than requiring structured search parameters.

Layer 2: Application Server - A Node.js/Express.js backend implementing: (1) AI Experience Weaver Service, (2) AI Travel Sync Service, (3) Mock Booking Engine. RESTful APIs abstract AI processing complexity.

Layer 3: Knowledge Base - A MongoDB NoSQL database serving as the semantic repository enriched with experiential tags—metadata describing abstract qualities of activities and destinations (e.g., peaceful, adventurous, romantic, budget_friendly, indoors).

Layer 4: Client-Side State Management - React state management handles real-time UI updates and coordinates API calls.

B. Core Innovation 1: AI Experience Weaver

The Experience Weaver implements a four-stage pipeline addressing discovery and generative composition:

1) Stage 1: NLP-Based Tag Extraction: User queries undergo syntactic parsing using compromise.js, a lightweight rule-based NLP library. The process: (1) Tokenization, (2) Part-of-Speech Tagging, (3) Tag Mapping against predefined experiential vocabulary (95 tags), (4) Entity Recognition for temporal constraints and locations.

Example: "I want a quiet, budget-friendly trip to the mountains for 3 days" yields [peaceful, budget_friendly] tags, [mountain] atmosphere, and 3 days duration. Empirical testing on 25 diverse test queries demonstrated approximately 88% tag extraction accuracy for straightforward queries containing 2-3 experiential adjectives.

2) Stage 2: Knowledge Graph Query: Extracted tags construct MongoDB queries implementing set-based semantic matching. The system applies ranking prioritizing entities with higher tag overlap, higher user ratings, and optimal budget alignment using the scoring function:

$$Score(e) = \alpha \cdot |T_e \cap T_q| + \beta \cdot R_e + \gamma \cdot B(e,b_q) \quad (1)$$

where T_e is entity tag set, T_q is query tag set, R_e is user rating, $B(e,b_q)$ is budget alignment function, and weights are empirically tuned ($\alpha = 3, \beta = 2, \gamma = 1$).

3) Stage 3: Itinerary Composition: The composition algorithm performs: (1) Day Segmentation across trip duration, (2) Time-of-Day Optimization (morning cultural/active → afternoon leisure → evening entertainment), (3) Geographical Clustering minimizing travel time, (4) Activity Type Balance ensuring variety, (5) Accommodation Selection matching experiential tags and optimal location. The system generates complete itineraries within 2.5-4.5 seconds.

C. Core Innovation 2: AI Travel Sync

The Travel Sync implements context-aware replanning through constraint-satisfaction:

1) Context Monitoring: During active trips, the system tracks current location, scheduled activities with timing, and activity environmental attributes (indoor/outdoor, duration).

2) Constraint Detection: Real-time contextual changes trigger conflict evaluation. Example: if weather = rain AND next activity = outdoor, the system triggers replanning through rulebased logic detecting environmental incompatibilities.

3) Alternative Search: Upon conflict detection, the system queries for alternatives satisfying new constraints while preserving original experiential intent tags. Response time for proactive suggestions is under 1.5 seconds.

D. Knowledge Base Design

The prototype Knowledge Base contains 110+ manually curated entities (60+ activities, 30+ accommodations, 20+ dining experiences) across 3 pilot destinations (Goa, Manali, Pune). Each entity is enriched with 8-12 experiential tags selected through firsthand research, review sentiment analysis from TripAdvisor, and expert validation to ensure high precision in semantic matching.

The tag ontology employs two tiers: (1) Emotional/Experiential attributes (peaceful, adventurous, romantic, cultural), (2) Practical constraints (indoors, outdoors, budget_friendly, family_friendly). Tags are non-mutually-exclusive; entities may possess multiple tags from both tiers enabling nuanced matching.

V. M ethodology

Algorithm 1 Experience-Led Itinerary Generation

[htbp]

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Input: *user query*, *duration*
Output: *structured itinerary*
PARSE: *tokens* ← *NLP.tokenize*(*user query*)
Extract *tags*, *location* from *tokens*
QUERY: *matching dest* ← *DB.find*({*tags* : {$*all* :

Algorithm 2 Context-Aware Dynamic Replanning

[htbp]

Input: *current itinerary*, *context change*
Output: *suggestion* or *null*
EVALUATE: *conflict* ← $detect\_violation(context, next\_activity)$
if *conflict* == *false* then
    return *null*
end if
SEARCH: *alt* ← *DB.find* ({ $tags$: {in : $new\_constraint, \$all : intent\}\})$
SUGGEST: Format *suggestion* with reasoning
return *suggestion* =0

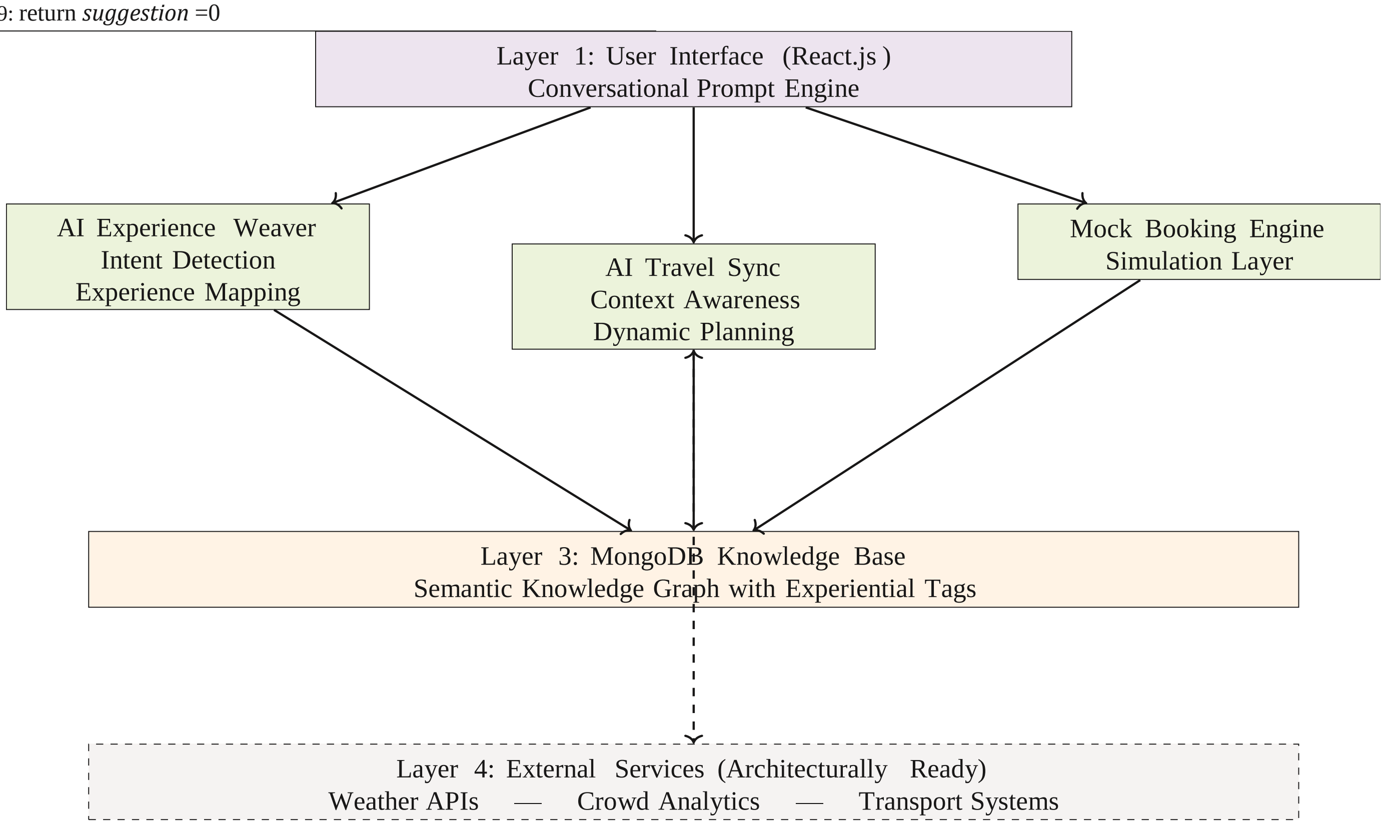

Fig. 1. Aura Travel System Architecture: Four-Layer AI-Driven Design

$tags\}\})$
RANK: $score \leftarrow tag\_overlap \times 3 + rating \times 2 + budget$
COMPOSE: Assemble *itinerary* with day segmentation
return *itinerary* =0

VI. Results and Evaluation

A. Implementation Status

The prototype followed three-phase agile development over 9-10 weeks: Phase 1 (MERN stack setup, mock booking engine), Phase 2 (AI Experience Weaver with Knowledge Base curation and NLP integration), Phase 3 (AI Travel Sync with simulation control panel).

B. Functional Validation

Table I presents test case results validating system capabilities.

TABLE I
Functional Validation Test Cases

Test Case	Result
Abstract Query	Input: “peaceful, budget-friendly mountains, 3 days”. Success: Generated Manali itinerary with nature walks, yoga, budget homestays. Tag accuracy: 100 %
Discovery	Input: “romantic beach escapes”. Success: Goa itinerary with beach resorts, sunset cruises. Relevance: 4.2/5
Replanning	Scenario: Beach scheduled, rain triggered. Success: Suggested Goa State Museum (indoors, cultural). Response: <2 s

Complex Query	Input: "adventurous safe family cultural trip". Success: Balanced itinerary with rafting (family), forts (cultural)

C. Performance Analysis and Failure Modes

While the system achieved 88% accuracy for straightforward 2-3 adjective queries, accuracy declined to approximately 65% for complex, multi-clause queries. This degradation reveals important technical insights:

Failure Mode 1: Ambiguous Negation - Queries containing negations ("not too expensive but also not cheap") confuse the rule-based parser, which cannot resolve implicit middle-ground constraints. The system incorrectly extracted both `budget_friendly` and `premium` tags, producing contradictory results.

Failure Mode 2: Temporal Complexity - Queries specifying time-variant preferences ("peaceful mornings, adventurous afternoons") exceed the current compositional algorithm's capabilities. The system treats all tags as uniform across the trip rather than time-segmented.

Failure Mode 3: Implicit Requirements - Abstract queries like "a trip my grandmother would enjoy" require demographic inference the system cannot perform. The lightweight NLP lacks semantic knowledge to map "grandmother" to tags like `accessible`, `gentle`, `cultural`.

These failure modes validate the architectural decision to use lightweight NLP for the prototype while highlighting the need for transformer-based models in production systems handling diverse query complexity.

D. Comparative Analysis

Table II presents multi-tiered comparison with existing platforms.

VII. Discussion

A. Architectural Design Decisions

Lightweight NLP vs. Transformers: The choice of compromise.js over transformer models represents deliberate optimization for prototype constraints. While transformers offer superior semantic understanding for complex queries (potentially addressing the 65% accuracy failure mode), they introduce GPU requirements, seconds-long inference latency, and need for domain-specific fine-tuning datasets. For proofof-concept validation, lightweight parsing provides sufficient demonstration while maintaining sub-5-second response times. Manual Curation vs. Automated Tagging: The Knowledge Base construction employed intensive manual curation ensuring tag precision. This approach is resource-intensive but provides high-quality ground truth validating the experiential matching hypothesis. Production requires hybrid approaches: automated baseline generation using sentiment analysis, expert validation, crowdsourced refinement.

B. Deployment Readiness

The system architecture is designed as ready for integration with external data sources:

Architectural Provisions:

- Abstract API interface layer supporting weather, crowd, and transportation data
- Modular service design enabling independent deployment and scaling
- Configuration-based external service registration Identified Integration Requirements:
- Weather: OpenWeatherMap or AccuWeather APIs ($0.0001/call)
- Crowd Monitoring: Google Popular Times API (free tier)
- Transportation: Google Maps Directions API ($0.005/request)
- Booking Integration: Affiliate APIs from major platforms

Scalability and Data Evolution: While the current prototype operates with 110+ entities across 3 pilot destinations, the MongoDB NoSQL architecture inherently supports horizontal scaling through sharding and replica sets. To transition to global scale (10,000+ entities), we propose a three-tier data acquisition pipeline: (1) automated baseline tag generation using sentiment analysis on review corpora (e.g., TripAdvisor), (2) expert validation by travel domain specialists, and (3) community-driven refinement through crowdsourced feedback. This hybrid approach ensures semantic precision is maintained as the knowledge base grows.

C. Ethical Considerations

Algorithmic Bias: Tag-based recommendation systems risk perpetuating biases present in underlying data. If the Knowledge Base overrepresents mainstream tourist attractions while underrepresenting off-the-beaten-path locations, the system systematically favors conventional tourism patterns, potentially accelerating overtourism and marginalizing local communities. Mitigation requires: (1) diverse, geographically distributed curation teams, (2) regular representation audits measuring destination and activity type distribution, (3) explicit mechanisms promoting lesser-known alternatives matching user intent.

Data Privacy: While the prototype does not store user data beyond session scope, production systems personalizing based on historical behavior must balance recommendation quality against privacy risks. Implementation must comply with GDPR, CCPA, and provide users transparency over data usage. The tag-based approach offers privacy advantages: personalization can occur based on current session context rather than extensive historical tracking.

User Agency: As AI systems become more sophisticated in suggesting and modifying itineraries, there's risk of reducing user agency and serendipity—core values in travel experience. The system must function as an intelligent assistant providing suggestions with clear reasoning and easy opt-out, not an autonomous decision-maker. Travel fundamentally involves

TABLE II
COMPARATIVE ANALYSIS OF TRAVEL PLANNING SYSTEMS

Feature	Traditional(Make-MyTrip)	AI Assistants (ChatGPT)	Aura Travel
Query Processing	Structured search required	Natural language, generic	NLP with travel-specific tags
Itinerary Generation	Manual selection	Generic suggestions	Complete bookable multi-day itineraries
Personalization	Collaborative filtering	Generic LLM	Semantic tag matching
Dynamic Adaptation	None (static)	Manual re-query	Automatedproactive replanning
Knowledge Source	Supplier databases	Web-scale data	Curated 110+ entities, 8-12 tags each
Real-Time Context	Not supported	Not supported	Architecturally ready (simulated)
Explainability	N/A	Low (black-box)	High (transparent tag matching)
Computational Cost	Low	Very High	Moderate (lightweight NLP)

human discovery; technology should enhance rather than replace exploration.

D. Limitations

Current limitations include: (1) Knowledge Base scope (3 destinations, 110 entities), (2) NLP accuracy ceiling for complex queries (65

VIII. FUTURE WORK

Future extensions would significantly enhance functionality for production deployment:

A. Advanced NLP Capabilities

Fine-tuning pre-trained transformer models (BERT, RoBERTa) on travel-specific dialogue datasets would address the 65% accuracy limitation for complex queries. Multi-turn dialogue management could iteratively refine ambiguous queries through clarifying questions ("By peaceful, do you prefer quiet nature or relaxed beach settings?").

B. Personalization Through Learning

Implementing privacy-preserving user profiles tracking preferences (accepted/rejected suggestions, explicit ratings) would enable collaborative filtering augmentation. Reinforcement learning agents could optimize replanning policies by learning which alternative types users prefer under different constraint scenarios, gradually improving suggestion quality.

C. Real-Time IoT Integration

Production deployment requires robust integration with weather forecasting (hourly granularity), crowd density monitoring (aggregated mobile location data), transportation status feeds (flight/train delays, traffic), and dynamic pricing services (real-time availability updates).

D. Multi-Modal Discovery

Extending discovery to support image-based queries ("I want a trip that looks like this photo") using computer vision embeddings (CLIP) matched against geo-tagged travel photo databases would enable new inspiration modes beyond text.

E. Group Trip Planning

Supporting multi-person trips requires preference aggregation algorithms finding itineraries satisfying multiple users' experiential intents with fairness constraints ensuring everyone's interests are represented.

F. Sustainability Integration

Calculating and displaying carbon footprints for itinerary choices, suggesting lower-impact alternatives, and implementing destination load balancing to prevent overtourism would address growing traveler concerns about sustainable tourism.

IX. CONCLUSION

This paper presented Aura Travel, an AI-powered platform addressing fundamental limitations in travel planning through hybrid architecture integrating NLP with semantically enriched Knowledge Graphs. The AI Experience Weaver enables experience-led discovery while the AI Travel Sync provides context-aware dynamic replanning. The prototype validates technical feasibility with 88% NLP accuracy for straightforward queries, 2.5-4.5 second itinerary generation, and sub-1.5 second replanning response times. Detailed failure analysis revealing 65% accuracy for complex queries provides technical insights into lightweight NLP limitations and validates the need for transformer-based approaches in production systems. The manually curated Knowledge Base of 110+ entities with 8-12 experiential tags enables semantic matching on abstract concepts that traditional systems cannot capture. Key contributions include the novel hybrid architecture, experiential tag ontology, dual-phase coverage of both discovery and execution, and empirical validation demonstrating both capabilities and limitations. This work demonstrates that thoughtful combination of established AI techniques with domain-specific innovation creates intelligent systems understanding abstract human intent and adapting dynamically to changing conditions.

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